



Diabetes: Type 1 & 2

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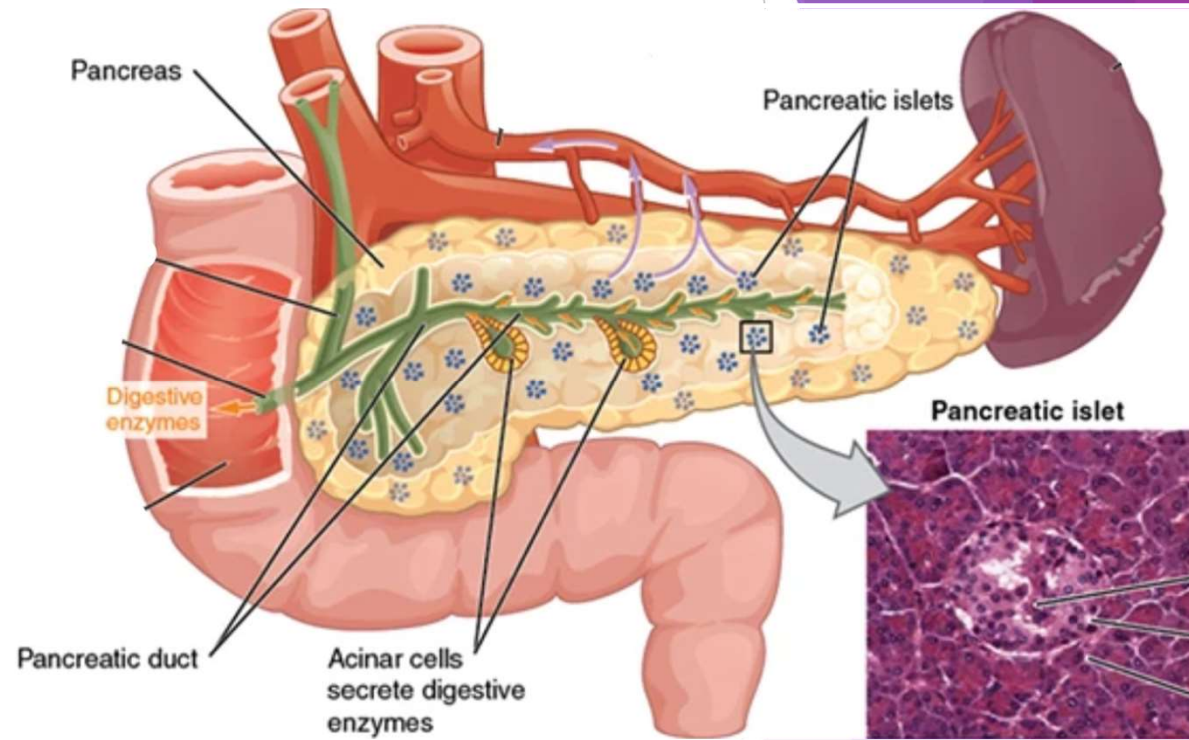
Glucose Metabolism: From Food to Fuel

- Carbohydrates are broken down into simple sugars for cells to use
- Digestion of carbohydrates starts with Amylase in the mouth
- As food moves along through the stomach to intestines, the pancreas releases insulin
- Insulin signals the liver to convert excess glucose into glycogen (stored in the liver) that can be used between meals/ if glucose levels drop
- When blood glucose decreases, the pancreas releases glucagon which signals the liver to break down glycogen stores back into glucose
- So, the pancreas and liver work together to keep balanced blood glucose for cells to utilize - especially the BRAIN who needs glucose to function



Pancreas, the Insulin Provider

- Insulin is a hormone made from beta cells inside the pancreas
- Works as a key to open the cell doors to Glucose and other nutrients
- The rise in blood glucose after a meal stimulates the pancreas to release Insulin.
- Type 1 Diabetes: Pancreas no longer makes insulin
- Type 2 : Pancreas makes some insulin, but the cells aren't as receptive to the insulin (insulin resistance)



What is Diabetes Mellitus?

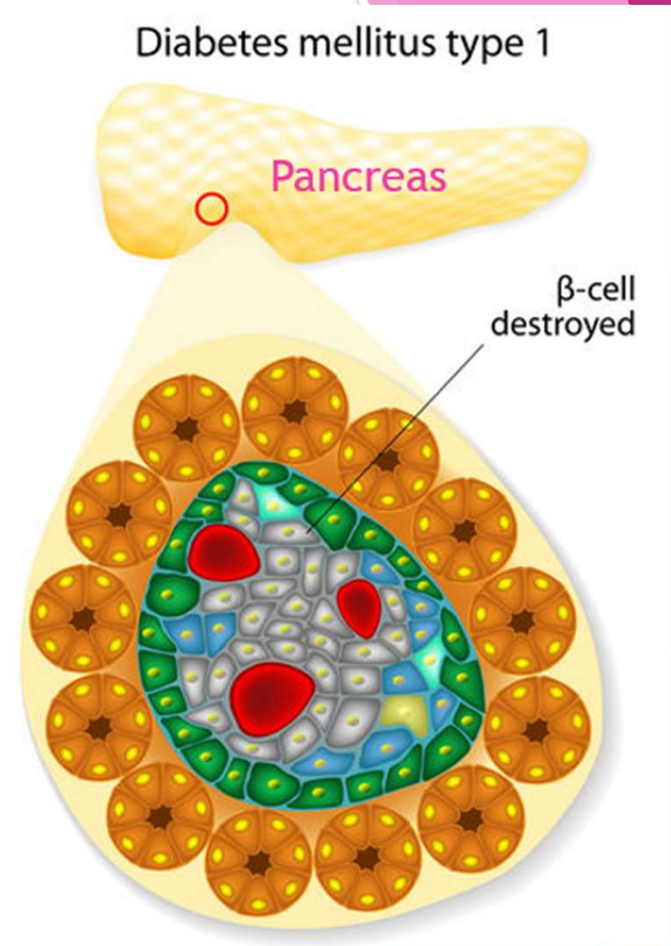
Chronic disease where glucose metabolism is impaired due to body's inability to produce insulin or use it effectively.

RISK FACTORS

- Age >30 years for Type 2 & <30 years for Type 1
- HDL $\leq$ 35 mg/dL and/or triglyceride level $\geq$ 250 mg/dL
- History of gestational diabetes or delivery of a baby over 9 lb
- Hypertension
- Family history of diabetes (parents or siblings with diabetes)
- Obesity
- Previously identified impaired fasting glucose

Type 1 DM

- Early in childhood, the immune system attacks and destroys insulin-producing beta cells in the pancreas
 - Not 100% established but theorized it can be caused by infection (Enteroviruses, Rotavirus, Cytomegalovirus), autoantibodies against beta cells, or genetics.
- Signs & Symptoms develop early in childhood or adolescence
 - S&S get worse as less insulin is produced
 - 3 Ps!! Polyurea, Polydipsia, Polyphagia (what's the logic?)
 - Weight Loss due to impaired carbohydrate metabolism
- Lifelong therapy including exogenous insulin
 - Very close management
 - BG Monitoring devices
 - Insulin pump

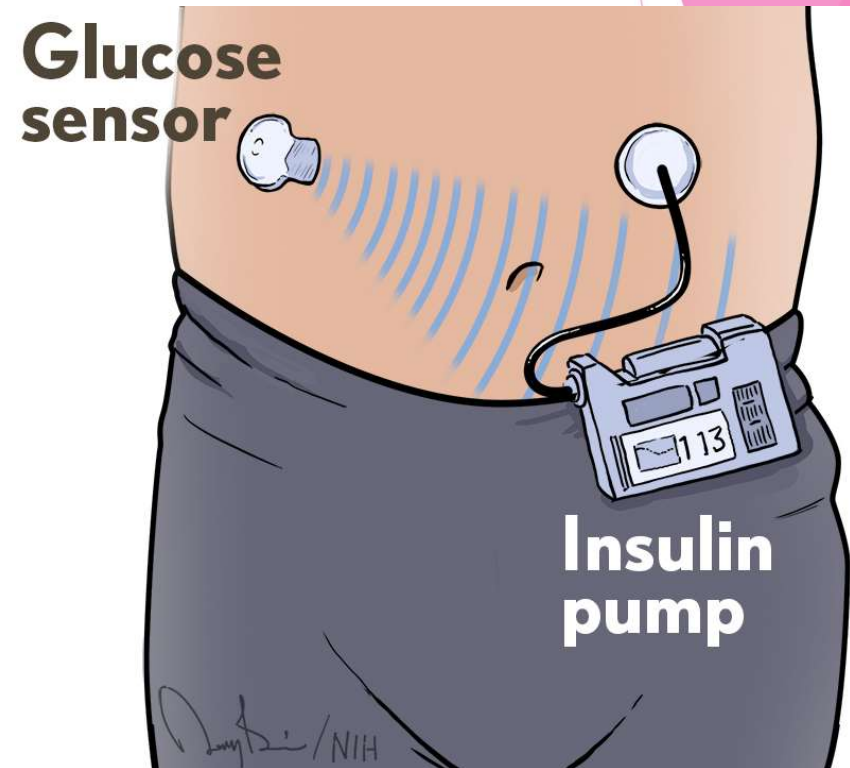


KEY POINT: consider the logic of glucose metabolism & how this would break down if beta cells are destroyed...

Type 1 DM Treatment

- **Exogenous Insulin**
 - Via Injections or continuous personal Insulin pump
- **Close, consistent, & daily blood glucose monitoring**
 - Via finger sticks or continuous personal BG monitor
- **Healthy lifestyle choices including Carbohydrate counting**

(Similar to Type 2, but must have exogenous Insulin!!)



Type 2 DM

- Inside the pancreas, beta cells make the hormone insulin.
- Beta cells release insulin to help the body use or store blood glucose from food
- The beta cells have been destroyed and insulin shots are the only way to keep blood glucose levels down.
- People with type 2 diabetes make insulin, but their bodies don't respond well to it.
- Only some people with type 2 diabetes need diabetes pills or insulin shots to help their bodies use glucose for energy.

TYPE 2 DM

- Cells in the body experience increasing insulin resistance - usually due to over-exposure to higher levels of glucose in the blood over time. This results in impaired glucose metabolism.
- Type 2 is the most common form of diabetes (DM).
- In this form, the body does not produce enough insulin or the cells become resistant to the insulin that is produced (can be both!)
- Treatment starts with diet & exercise
- Then add PO medication (usually metformin)
- ...then add more PO medication (dual therapy)
-then add Insulin or other injectable treatment

(We'll review pharmacology in another PowerPoint!)

KEY POINT: consider the logic of glucose metabolism & how this would break down if beta cells' insulin production is impaired & if cell receptors are impaired...

What is a normal blood sugar?

70-110mg/dL

This answer varies from lab to lab, book to book.

Go with what's in your book. 😊

Diagnosis Criteria

Chart 46-2

Criteria for the Diagnosis of Diabetes

- Symptoms of diabetes plus casual plasma glucose concentration equal to or greater than 200 mg/dL (11.1 mmol/L). Casual is defined as any time of day without regard to time since last meal. The classic symptoms of diabetes include polyuria, polydipsia, and unexplained weight loss.

Or

- Fasting plasma glucose greater than or equal to 126 mg/dL (7.0 mmol/L). Fasting is defined as no caloric intake for at least 8 hours.

Or

- Two-hour postload glucose equal to or greater than 200 mg/dL (11.1 mmol/L) during an oral glucose tolerance test. The test should use a glucose load containing the equivalent of 75-g anhydrous glucose dissolved in water.

Or

- Hemoglobin A1C $\geq 6.5\%$ (48 mmol/mol).

In the absence of unequivocal hyperglycemia with acute metabolic decompensation, these criteria should be confirmed by repeat testing on a different day. The third measure is not recommended for routine clinical use.

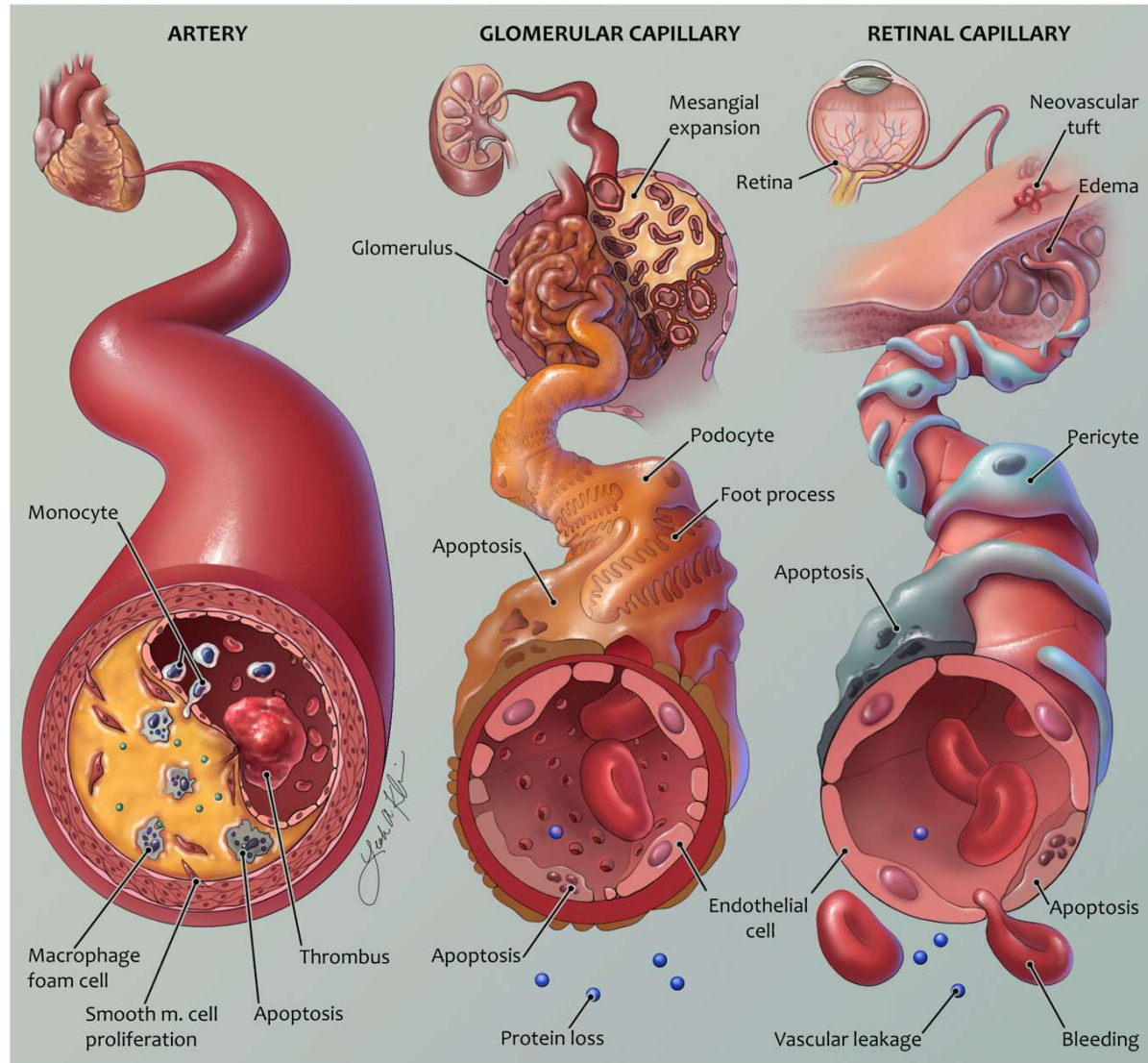
A1C, glycosylated hemoglobin

Adapted from American Diabetes Association (ADA). (2020). Standards of medical care in diabetes—2020. *Diabetes Care*, 43(Suppl 1), S1–S212.

Risks in uncontrolled Diabetes

- Neuropathy - affects half of diabetics
- Skin Complications
- Foot wounds
- Increased risk for infection
- Oral Complications including gingivitis & periodontitis
- Diabetes-Related Eye Disease
- Hearing loss (twice as common in diabetics, 30% higher in pre-diabetics)
- Chronic Kidney Disease
- Cardiovascular Disease - leading cause of death for diabetics
- Stroke - Risk is doubled for diabetics

MACRO vs. MICRO MICRO

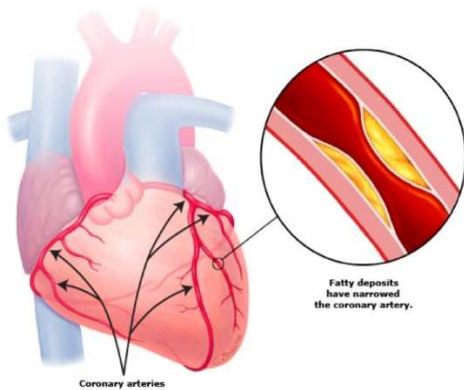


MACRO-VASCULAR VS MICRO-VASCULAR CHANGES & COMPLICATIONS

Microvascular & Macrovascular Changes

Macrovascular Changes

- Cardiovascular Disease
- Peripheral Artery Disease

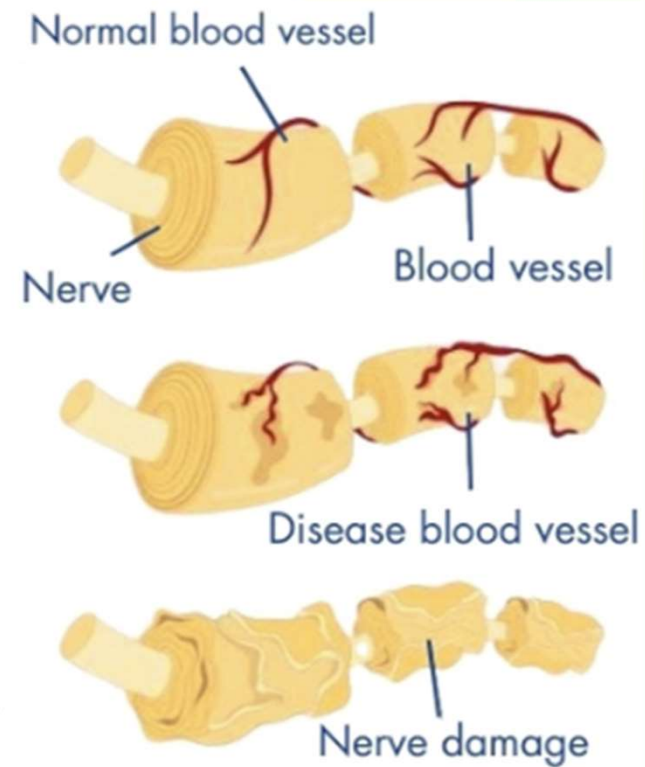


Microvascular Changes

- Neuropathy
- Nephropathy
- Retinopathy

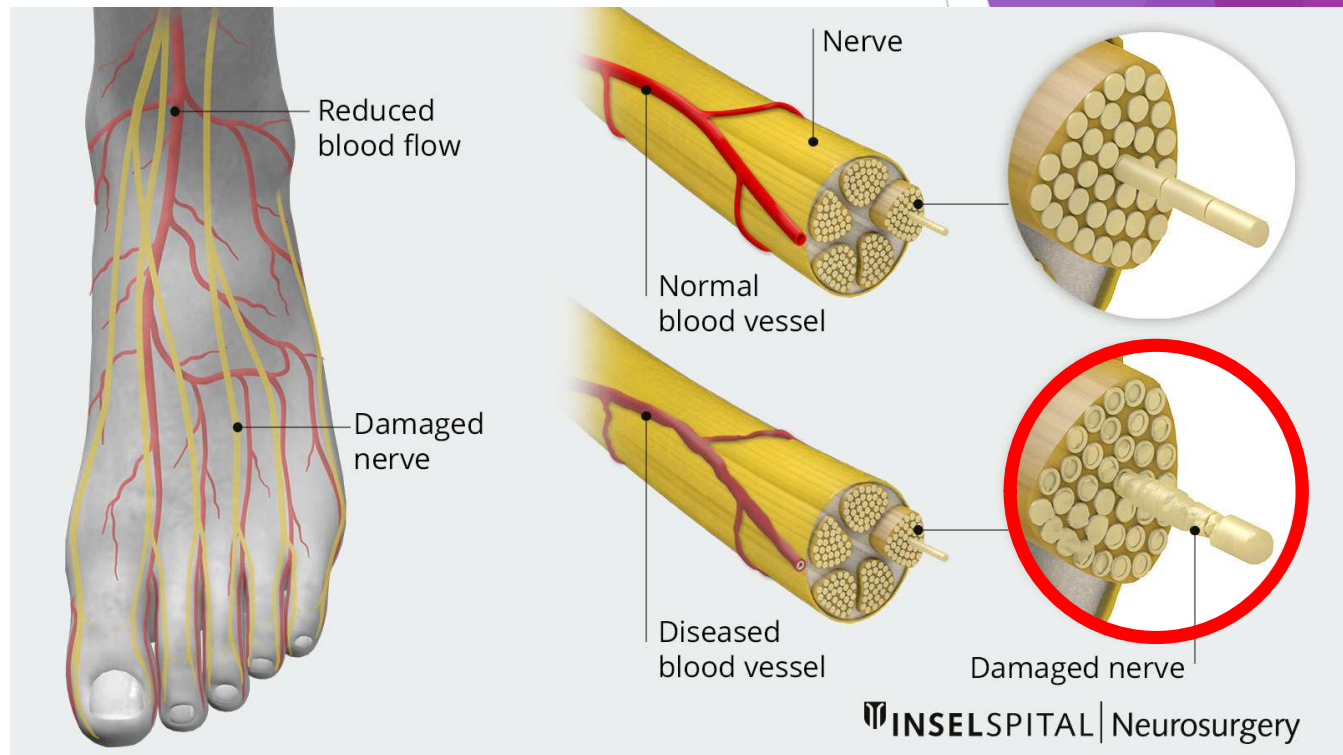
Microvascular:

Think of the
smallest & most
distal vessels &
nerves →



Neuropathy & Foot Complications

- Nerves become damaged
- Sensation is impaired
- MicroCirculation is impaired
- Wounds develop & healing is delayed
- Can lead to amputation
 - Toe(s)
 - Partial/Total Foot
 - Below knee (BKA)
 - Above knee (AKA)



What do we manage as nurses?

- Everything we already talked about... and more acutely:
 - Hypoglycemia
 - Hyperglycemia
 - Diabetic Ketoacidosis (DKA)
 - Hyperosmolar Hyperglycemic Nonketotic Syndrome (HHNS)

(DKA & HHNS will be covered in 4th Semester in great detail!!)

HYPOGLYCEMIA & HYPERGLYCEMIA

BLOOD SUGAR MNEMONIC

HOT & DRY = SUGAR HIGH



COLD & CLAMMY =
NEED SOME CANDY

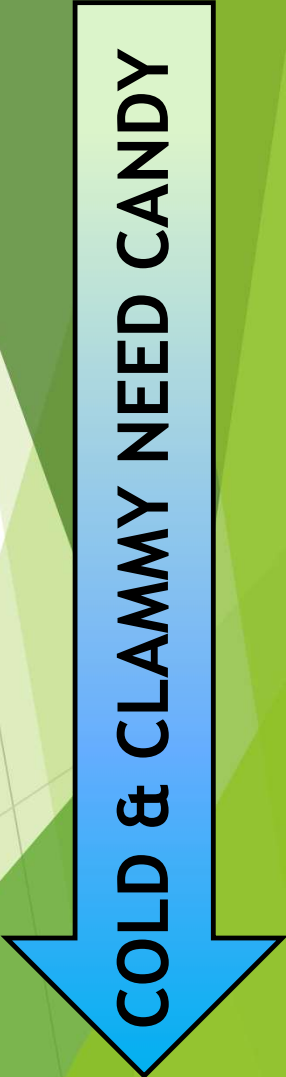
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- The body can handle having HIGH Blood Glucose better than having LOW Blood Glucose
- Either extreme can be an urgent or emergent issue!

HYPOGLYCEMIA

- Blood Glucose < 70 mg/dl (Some facilities/labs use < 80mg/dl)
- Symptoms:
 - Sweating - “cool & clammy”
 - Shakiness, feeling “jittery”
 - Rapid HR
 - Confusion
 - Weakness
 - Headache

always depends on the patient...



COLD & CLAMMY NEED CANDY

15-15 Rule for Treating Hypoglycemia

15-15 Rule

① TEST BLOOD GLUCOSE



② Eat/Drink 15 GRAMS of FAST-ACTING CARBS



③ WAIT 15 MINUTES



*REPEAT UNTIL BLOOD SUGAR LEVELS ARE >70mg/dl

COLD & CLAMMY NEED CANDY

Give Simple Carbs until BG >70mg/dl THEN Complex Carbs

HYPERGLYCEMIA

- Elevated blood glucose >126 mg/dl
- Symptoms:
 - Polyphagia
 - Polydipsia
 - Polyuria
 - Dry mouth
 - Blurred vision
 - Frequent headaches

HOT & DRY NEED SOME “I”

“I” = Insulin!!

Patient Education

- How to check their blood sugar
 - Using Glucometer & Recording results in a log
- How to read the results
- Signs & Symptoms of Hypo- & Hyper- Glycemia & when to check it
- Interventions for High/Low & when to call their provider
- Sick Day Rules
 - Stay hydrated
 - Monitor BG more frequently (every 4-6 hours)
 - Continue taking diabetic medications
 - Contact provider with complications / persistent hyperglycemia

KETONES & DIABETIC KETOACIDOSIS (DKA)

- Ketones are acids that can build up in the blood
- When the body doesn't have enough insulin to break down carbohydrates, it breaks down fat & protein to use for energy & ketones are a byproduct that appears in urine

Type 1 Diabetics should check urine for ketones if:

- Blood Glucose >240mg/dL
- Any unusual symptoms
- Illness or Infection present

(Changes may indicate a change in Insulin needs)

High amounts of Ketones may lead to Diabetic Ketoacidosis

- Usually Type 1 Diabetes
- Also in Type 2 but usually with other illness present

KETOACIDOSIS

Pathophysiology:

- Lack of Insulin from Pancreas results in impaired carbohydrate metabolism
- Fat & Protein broken down for energy (mostly fat - “lipolysis”)
- Liver converts fatty acids into Ketones via “ketogenesis”

Warning signs of DKA:

- Thirst or a very dry mouth
- Dry or flushed skin
- Polyuria & Polydipsia
- Hyperglycemia
- High levels of ketones in the urine
- Persistent fatigue
- Nausea, vomiting, or ABDOMINAL PAIN
- Fruity odor on breath
- A hard time paying attention or confusion
- Dyspnea (short, deep breaths - Kussmaul's Respirations - BUT WHY??)

KETOACIDOSIS TREATMENT

Three Main Clinical Features:

- Hyperglycemia
- Acidosis
- Fluid & Electrolyte Imbalance
 - Dehydration
 - Electrolyte Loss

Treatment:

- Correct Electrolyte deficits
- IV Regular Insulin



Insulin Waning - Symogi Effect - Dawn Phenomenon

Insulin Waning (decreased effectiveness)

BG progressively rises through the night as insulin effectiveness *wanes* or wears off. TX: increasing evening (predinner or bedtime) dose of intermediate or long-acting insulin or adding a dose of insulin before the evening meal.

Progressive $\uparrow$ BG $\uparrow$
thru the night

Dawn Phenomenon (hormonal effect)

Hormones such as growth hormone, cortisol, and glucagon increase in early am leading to insulin resistance and higher blood glucose in the early am. TX: changing time of injection of evening intermediate-acting insulin from dinnertime to bedtime.

$\uparrow$ BG $\uparrow$ in am just
before/after waking

Somogyi Effect (rebound hypERGlycemia)

Normal or elevated blood glucose at bedtime, early morning hypoglycemia, and rebound hyperglycemia caused by the production of counter-regulatory hormones. TX: decreasing evening (predinner or bedtime) dose of intermediate-acting insulin, or increasing bedtime snack.

Early am $\downarrow$ BG $\downarrow$ then
rebound $\uparrow$ BG $\uparrow$ in am

***Note: Remember nurses do not adjust meds, we collab with providers who adjust orders!
But we CAN check BG through the night to see the trends & identify these issues!***